

# Lab Manual: Deploying a Streamlit App to Streamlit Cloud

## Objective

By the end of this lab, you will:

- Prepare your Streamlit project for deployment.
- Push your code to GitHub.
- Deploy your app on **Streamlit Cloud**.
- Get a **public URL** to share your app with others.

## Prerequisites

- A working **Streamlit app** (app.py or streamlit\_app.py).
- Python installed locally.
- **Git** installed.
- A **GitHub account**.

## Step 1: Prepare Your Streamlit Project

1. Ensure your main file is named: app.py (or streamlit\_app.py).
2. Create a requirements.txt with all dependencies: `pip freeze > requirements.txt`

This tells Streamlit Cloud what packages to install.

## Step 2: Push Code to GitHub

Run the following commands in your project folder:

```
# Initialize Git
git init

# Add all files
git add .

# Commit changes
git commit -m "Initial commit - Streamlit app"

# Add remote (replace with your repo link)
git remote add origin https://github.com/your-username/your-repo.git

# Push to GitHub
git push -u origin main
```

Your code is now stored in a GitHub repository.

## Step 3: Deploy on Streamlit Cloud

1. Go to  <https://share.streamlit.io>.
2. Sign in with your **GitHub account**.
3. Click **New App**.
4. Select:
  - a. Repository: your repo name.
  - b. Branch: usually main.
  - c. File path: app.py.
5. Click **Deploy**.

Streamlit Cloud will:

- Install packages from requirements.txt.
- Run your app automatically.

## Step 4: Access Your Public App

- After a few minutes, you'll get a **unique URL** like:

<https://your-username-your-repo.streamlit.app>

- Open the link to test your app.
- Try uploading files, using widgets, and verifying functionality.

## Step 5: Make Updates

Whenever you change your app locally:

```
git add .  
git commit -m "Update app features"  
git push
```

Streamlit Cloud automatically redeploys the latest version.

## Verification Checklist

- `app.py` exists as the main file.
- `requirements.txt` created with dependencies.
- Code pushed to GitHub successfully.
- App deployed via Streamlit Cloud.
- Public URL generated and app works.
- Updates redeploy automatically after pushing changes.

## Summary

In this lab, you:

- Prepared your app with `requirements.txt`.
- Uploaded code to GitHub.
- Deployed on **Streamlit Cloud** in just a few clicks.
- Shared a **public URL** with others.